

Cervical Spine MRI

ACG: A-0057 (AC)

[Link to Codes](#)

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Clinical Indications

- Cervical spine MRI may be indicated for **1 or more** of the following(1)(2)(3)(4)(5):
 - ☐ Cancer or neoplasm evaluation, staging, or surveillance needed, as indicated by **1 or more** of the following(20)(21)(22)(23)(24)(25):[\[1\]](#)
 - Bone neoplasm (benign or malignant) involving spine, known or suspected (eg, chondrosarcoma, chordoma, Ewing sarcoma family of tumors, giant cell tumor of bone, osteosarcoma, metastatic tumors), as indicated by **1 or more** of the following(32)(33):
 - Initial evaluation or staging(34)
 - Monitoring disease progression or response to treatment
 - Post-treatment surveillance
 - Histiocytic or dendritic cell neoplasm^[A] with involvement of vertebrae, known or suspected(35)(38)(39)
 - Localized neck pain and **1 or more** of the following(40)(41):
 - Bone scan positive(34)
 - Pain occurs mainly at night.(30)(42)
 - Persistent pain in patient older than 50 years(43)
 - Personal history or concurrent diagnosis of malignancy(44)(45)
 - Weakness, rapidly progressing(43)
 - Weight loss, unexplained(42)(43)
 - Myelopathy signs or symptoms (eg, motor weakness, bowel or bladder dysfunction)(46)(47)(48)
 - Primary central nervous system tumor (eg, adult-type or pediatric-type diffuse glioma, ependymoma, medulloblastoma, other central nervous system embryonal tumor, primary central nervous system lymphoma, meningioma), and need for **1 or more** of the following(33)(49)(50)(51)(52)(53):
 - Brain tumor, and need for evaluation for spinal involvement
 - Initial evaluation, staging, or treatment planning
 - Monitoring disease progression or response to treatment
 - Post-treatment surveillance
 - Primary non-central nervous system tumor, and need for diagnosis, staging, or monitoring of known or suspected spinal cord, spine, or leptomeningeal metastasis, as indicated by **1 or more** of the following(33)(49)(52)(53)(54):
 - Known spinal cord, spine, or leptomeningeal metastasis, and need for monitoring disease progression or response to treatment
 - Post-treatment surveillance

- Suspected spinal cord, spine, or leptomeningeal metastasis (eg, due to myelopathy, pain, finding on other imaging), and need for evaluation, staging, or treatment planning
 - Screening for central nervous system tumor in patient with high-risk genetic disorder or syndrome (eg, neurofibromatosis type 2, von Hippel-Lindau syndrome)(53)(55)(56)(57)(58)
 - Cerebrospinal fluid leak, suspected(20)(59)(60):[N](#)
- ☐ Chiari malformation, suspected, as indicated by **1 or more** of the following(20)(61)(62):[N](#)
 - Babinski sign positive(64)(65)
 - Gait disturbance(66)
 - Hyperreflexia(67)
 - Lower cranial nerve dysfunction such as vocal cord paralysis(66)(68)
 - Nystagmus(66)
 - Paroxysmal headache or posterior cervical pain with Valsalva maneuvers (eg, laughing, sneezing, coughing)(68)(69)
- ☐ Congenital spinal conditions, suspected, as indicated by **1 or more** of the following(20)(70)(71):[N](#)
 - Diastematomyelia (ie, split cord malformation)(72)
 - Dorsal dermal sinus(68)
 - Foramen magnum stenosis (eg, in patient with achondroplasia)(73)
 - Hemangioma(74)(75)(76)(77)(78)
 - Hydromyelia
 - Intradural lipoma
 - Myelocele
 - Myelomeningocele(72)
- ☐ Infection, known or suspected (eg, vertebral osteomyelitis, disk space infection, epidural abscess), as indicated by **ALL** of the following(20)(30)(79)(80)(81)(82):[N](#)
 - Risk factors for spinal infection, as indicated by **1 or more** of the following:
 - Bone scan or plain x-ray suggestive for infection(89)
 - C-reactive protein or erythrocyte sedimentation rate elevated(90)(91)(92)
 - Fever(48)(65)
 - History of intravenous drug abuse, other source of infection, or recent invasive procedure(93)
 - Immunosuppression(44)
 - Positive blood culture(90)(94)
 - Recent history of spinal surgery(95)
 - Tuberculosis, concurrent or suspected(44)(88)(96)(97)
 - Suspected spinal infection (eg, localized midline back pain, limp, refusal to walk)(65)(90)
- ☐ Inflammatory or demyelinating process, suspected (eg, clinically isolated syndrome, multiple sclerosis, other demyelinating disease), as indicated by **1 or more** of the following(20)(98)(99):[N](#)
 - Multiple sclerosis, suspected, as indicated by **1 or more** of the following(100)(101)(104)(110):
 - Ascending numbness or tingling (eg, from foot to trunk)(103)
 - Brown-Sequard syndrome(103)
 - Conditions mimicking multiple sclerosis (eg, Sjogren syndrome, systemic lupus erythematosus, antiphospholipid syndrome) cannot yet be excluded.(47)(111)(112)
 - MRI of brain nondiagnostic for multiple sclerosis(113)
 - Signs or symptoms of myelopathy or myelitis(48)(114)(115)
 - Transverse myelitis in spinal cord, suspected (idiopathic or in conjunction with multiple sclerosis), as indicated by **1 or more** of the following(44)(116)(117)(118):
 - Bilateral signs or symptoms of involvement of appropriate level of spinal cord
 - Clearly defined sensory level
 - Sudden onset of sensory, motor, and autonomic dysfunction attributable to appropriate level of spinal cord
 - Klippel-Feil syndrome, suspected (ie, fusion of 2 cervical vertebrae)(119)(120)(121)(122):[N](#)
- ☐ Pain localized to neck or radicular in nature, subacute or chronic, as indicated by **1 or more** of the following(41)(48)(126)(127)(128):[N](#)
 - Adult and **ALL** of the following:
 - Failure to improve after 6 or more weeks of nonoperative treatment, as indicated by **1 or more** of the following(48)(81)(129)(130)(131)(132)(133):
 - Activity modification(134)
 - Exercise(135)
 - Pharmacotherapy (eg, analgesics, anti-inflammatory medications)(136)(137)
 - Physical therapy(137)(138)(139)
 - Patient being considered for invasive treatment (eg, epidural steroids, surgery)
 - Significant interference with daily function
 - Child or adolescent and **ALL** of the following(30):
 - Age younger than 18 years

- Concern for serious underlying pathology (eg, malignancy, infection, inflammatory process), as indicated by **1 or more** of the following:
 - Associated signs suggestive of cancer (eg, easy bruising, abdominal mass, lymphadenopathy)
 - Fever
 - Gait abnormalities or refusal to walk or crawl
 - Morning stiffness
 - Neurologic deficit
 - Nocturnal pain or constant pain
 - Pain lasting more than 4 weeks or worsening pain
 - Unintentional weight loss
- ☐ Postoperative spinal complications, known or suspected, as indicated by **1 or more** of the following(140)(141)(142):^N
 - Postoperative hemorrhage or hematoma(95)
 - Significant new neurologic findings (eg, spinal cord impingement)
 - Significant new pain symptoms (eg, recurrent or new radicular symptoms, delayed pain at fusion site)(41)
- ☐ Rheumatoid arthritis in cervical spine with suspected instability or cord compression, as indicated by **1 or more** of the following(118)(143)(144)(145):^N
 - Atlantoaxial subluxation or impaction on plain x-ray
 - Evidence of myelopathy or progressive neurologic deficit
 - Subluxation of first and second cervical vertebrae (C1 and C2)
- ☐ Scoliosis, as indicated by **1 or more** of the following(20)(147)(148)(149):^N
 - Congenital scoliosis
 - Early-onset scoliosis (age 9 years or younger)
 - Neurofibromatosis
 - Presurgical planning for adolescent idiopathic scoliosis to assess possible neural axis malformation, as indicated by **1 or more** of the following(147)(149)(151)(153):
 - Abnormal neurologic findings on clinical examination(154)
 - Age at first visit 10 years or younger
 - Kyphosis at curve apex
 - Left-sided thoracic curvature
 - Male gender
 - Pain, moderate to severe(6)(154)
 - Rapid curve progression (ie, more than 1 degree per month)
 - Short segment curve (ie, less than 6 vertebral segments)
 - Thoracic kyphosis 30 degrees or greater
 - Vertebral abnormalities (eg, hemivertebrae, block vertebrae) detected on x-ray
- Spinal cord compression, myelopathy, or cauda equina syndrome, suspected (eg, positive Babinski sign, depressed ankle deep tendon reflexes, gait abnormality, hyperreflexia, sensory or motor deficits)(20)(27)(44)(155)(156)(157)^N
- Spinal cord infarction or ischemia, suspected (eg, sudden-onset back pain, sensory or motor deficits, urinary retention)(159)^N
- Spinal cord stimulator placement planning needed, and no MRI of region within 12 months prior to placement(164)(165)^N
- ☐ Spinal dysraphism, suspected, as indicated by **1 or more** of the following(63)(72)(169)(170)(171)(172)(173):^N
 - For infant, as indicated by **1 or more** of the following(175)(176)(177)(178):
 - Absent reflexes
 - Aplasia cutis congenita
 - Decreased spontaneous leg movement
 - Decreased urinary stream
 - Deviation of gluteal cleft
 - Foot asymmetry(68)
 - Leg atrophy
 - Midline lumbosacral skin lesions (eg, atypical dimple, dermal sinus, lipoma, tail)(6)(68)
 - Unilateral lumbosacral hypertrichosis
 - For toddler: developmental delay in walking, abnormal gait
 - For older child: spasticity, hyperreflexia, bowel or bladder disturbance, including incontinence(68)
- ☐ Spinal stenosis of cervical spine, suspected, as indicated by **ALL** of the following(20)(155)(179):^N
 - Patient being considered for invasive treatment(181)
 - Progressive or disabling symptoms of cervical spinal stenosis, as indicated by **1 or more** of the following(180):
 - Hyperactive reflexes
 - Muscle weakness
 - Sensory loss
 - Spasticity(182)
- Spondylolysis, known or suspected (eg, plain radiography-negative early spondylolysis, radiculopathy in patient with spondylolysis)(30)(183)(184)(185)^N
- Stereotactic spine radiotherapy treatment planning(187)(188)(189)(190)(191)^N

- ☐ Syringomyelia in cervical spine, suspected, as indicated by **1 or more** of the following(20)(66)(194)(195):^[1]
 - Upper limb sensory loss in cape-like distribution(61)
 - Weakness, muscle wasting, and sensory loss beginning in intrinsic muscles of hand
- ☐ Torticollis, as indicated by **ALL** of the following(122)(196)(197):^[1]
 - Age younger than 18 years
 - Initial CT scan normal or inconclusive
 - Nontraumatic etiology
- ☐ Trauma or fracture of cervical spine, known or suspected, as indicated by **1 or more** of the following(198)(199)(200)(201):^[1]
 - Age 65 years or older(218)
 - Altered mental status (204)(213)(219)(220)
 - Child younger than 3 years with acute head trauma, and physical abuse suspected(214)
 - Epidural hematoma, suspected(206)(221)
 - Fall from height greater than 3.3 feet (1 meter)(218)
 - High clinical suspicion of acute spinal injury, as indicated by **ALL** of the following(220)(222)(223)(224)(225):
 - Negative or indeterminate cervical spine x-ray or CT scan results(213)(218)(226)
 - Symptoms, as indicated by **1 or more** of the following(227):
 - Midline cervical tenderness
 - Paresthesias of extremities
 - Ligamentous injury, suspected(196)(217)(220)(228)
 - Multiple traumatic injuries (eg, concurrent injury to 2 or more body parts or systems)(229)
 - Neurologic symptoms associated with traumatic mechanism of injury, as indicated by **1 or more** of the following(20)(218)(228)(230)(231)(232):
 - Ejection from motorcycle(227)(233)
 - High-speed motor vehicle accident (greater than 50 mph (80.5 kph))(227)(233)
 - Intoxicated state(234)
 - Minimal trauma in osteoporotic patient with suspected fragility fracture
 - Motor vehicle accident, with acute hyperflexion of spine over seat belt(235)
 - Pediatric patient(215)(218)(236)(237)(238)
 - Results of plain x-ray or CT scan nondiagnostic or show abnormal spine motion or burst fracture(207)(208)(239)(240)
 - Traumatic disk herniation, suspected(228)
- ☐ Tuberculosis of spine, known or suspected, as indicated by **1 or more** of the following(88)(121)(241)(242):^[1]
 - Diagnostic evaluation of suspected spinal tuberculosis
 - Preparation for surgical management
- ☐ Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
 - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

Some gadolinium-based contrast agents used for diagnostic and treatment guidance with MRI or MRA have been associated with nephrogenic systemic fibrosis in patients with chronic kidney disease or acute kidney injury.(6)(7)(8)(9)(10) (**EG 2**) Gadolinium deposition in the brain has been reported by multiple research groups; however, the clinical significance of this finding is unclear.(6)(7)(11)(12) (**EG 2**) Specialty society guidelines recommend weighing the risks vs benefits when using gadolinium-based contrast agents for MRI.(13) (**EG 2**) MRI-based techniques may be contraindicated in patients with ferromagnetic intracranial aneurysm clips and certain types of ferromagnetic foreign bodies and cochlear implants.(14) (**EG 2**) Although these implants have traditionally included cardiac pacemakers and implantable cardioverter-defibrillators, multiple trials have validated the safe and effective use of pacemaker systems designed for MRI use, and a guideline supported by multiple specialty societies states that many patients with implantable cardiac devices, including those not originally intended for MRI scanning, can safely undergo MRI scanning under defined clinical indications.(7)(15)(16)(17)(18)(19) (**EG 1**)

Criteria

The evidence for the clinical indications found in this guideline includes 202 published peer reviewed articles, 10 specialty society or other evidence-based guidelines, and 13 book sections.

For cancer or neoplasm evaluation or staging, evidence demonstrates at least moderate certainty of at least moderate net benefit. (**RG A1**) Whole-spine MRI is the preferred imaging modality for suspected metastatic disease to the spine in patients with known primary

malignancy; metastatic epidural spinal cord compression is seen in 5% to 10% of patients with cancer.(26)(27) **(EG 2)** Spinal cord tumor in children may present with persistent localized back pain that is worse at night.(28)(29)(30) **(EG 2)** Additional findings in children with spinal cord tumor may include burning pain with paresthesia, gait disturbance, paraspinal muscle spasm, progressive motor weakness, progressive scoliosis, and sensory deficits; cervical spine lesions may cause obstruction of the foramen magnum leading to symptoms of increased intracranial pressure.(31) **(EG 2)**

For cerebrospinal fluid leak, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline notes that MRI of the spine may be useful to discover the location of a cerebrospinal fluid leak in patients with intracranial hypotension.(20) **(EG 2)**

For Chiari malformation, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Cervical spine MRI is indicated to diagnose miscellaneous congenital spinal deformities, including Chiari malformation and spinal dysraphisms.(61)(63) **(EG 2)**

For congenital spinal conditions, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline indicates that MRI is an appropriate imaging modality for assessment of congenital spinal conditions.(20) **(EG 2)**

For infection of the spine, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRI is a preferred option for diagnosis of suspected spinal infection and inflammation, including spondylitis, epidural abscess, osteomyelitis, and spinal tuberculosis.(83)(84)(85)(86)(87) **(EG 2)** Because spinal tuberculosis can present in noncontiguous vertebral bodies, patients with known spinal tuberculosis should undergo MRI imaging of the entire spine.(88) **(EG 2)**

For inflammatory or demyelinating processes, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Diagnostic criteria have been developed for inflammatory or demyelinating processes, such as clinically isolated syndrome or multiple sclerosis (most commonly manifested as unilateral optic neuritis), brainstem syndrome, or partial myelitis, based on the demonstration of central nervous system lesions disseminated in space and disseminated in time on MRI imaging of the brain and, if appropriate, on spinal cord imaging.(100)(101)(102)(103)(104)(105) **(EG 2)** Spinal cord involvement is seen on MRI imaging in 80% to 90% of patients with multiple sclerosis; asymptomatic spinal cord lesions may be present in up to 40% of patients with clinically isolated syndrome, with lesions occurring more commonly in the cervical spine than in the thoracic spine.(106)(107)(108)(109) **(EG 2)** An expert consensus panel recommends spinal cord imaging for patients with suspected demyelinating disease who present with signs or symptoms of spinal cord involvement or if the number of lesions on brain MRI is insufficient to meet established criteria for multiple sclerosis.(100) **(EG 2)**

For Klippel-Feil syndrome, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Klippel-Feil syndrome can predispose to cervical spondylotic myelopathy later in life.(119)(123)(124) **(EG 2)** Performing cervical spine MRI and CT scan in Klippel-Feil syndrome and other congenital cervical anomalies may help to identify the presence and degree of craniovertebral dislocation and atlantoaxial rotatory subluxation, which may present as acute spinal cord compression in children.(120)(121)(125) **(EG 2)**

For pain localized to the neck or radicular in nature, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Review articles consider MRI an appropriate initial imaging modality for pain that significantly interferes with daily function and for which invasive treatment is being considered.(126)(127)(128) **(EG 2)** A specialty society guideline states that MRI may be appropriate for the evaluation of acute or increasing cervical pain with radiculopathy and for chronic cervical pain.(41) **(EG 2)** For children and adolescents with back or neck pain, a specialty society guideline notes that, while imaging is not generally indicated for pain that is self-limited or not associated with signs or symptoms of an underlying disorder, MRI is appropriate for evaluating pain associated with potential red flags (eg, associated gait abnormalities, unintentional weight loss, persistence over more than 4 weeks).(30) **(EG 2)**

For postoperative spinal complications, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles indicate that MRI is effective at demonstrating common postoperative complications after spinal surgery.(140)(141)(142) **(EG 2)**

For rheumatoid arthritis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** The prevalence of cervical spine involvement in rheumatoid arthritis ranges from 16% to 70%, presenting as atlantoaxial subluxation, subaxial subluxation, or cranial settling.(146) **(EG 1)** MRI is the most useful test for patients with signs and symptoms suggestive of cervical spondylotic myelopathy; however, if MRI is contraindicated, CT myelography is recommended.(64)(143) **(EG 2)**

For scoliosis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles conclude that MRI is not routinely indicated for pain in the absence of abnormal neurologic examination findings or an atypical curve, such as a left-sided curve (levoscoliosis), a short segment, or kyphosis near the curve apex.(149)(150)(151)(152) **(EG 2)** However, it is useful preoperatively in evaluating the neural axis.(153) **(EG 2)**

For spinal cord compression, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review concluded that MRI is appropriate for diagnostic confirmation of cervical spondylotic myelopathy.(158) **(EG 2)** MRI is the preferred imaging modality for detection and confirmation of malignant spinal cord compression.(46) **(EG 2)** Literature reviews of acute transverse myelitis recommend urgent neuroimaging with MRI to rule out spinal cord compression; however, if MRI is not available or is contraindicated, CT myelography is indicated.(99)(116) **(EG 2)**

For spinal cord infarction or ischemia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** MRI of the spine is the preferred diagnostic procedure to detect spinal cord ischemia. The clinical manifestations may evolve over minutes to hours, vary depending on the vascular territory involved, may be bilateral or unilateral, and may present transiently (also called spinal transient ischemic attacks).(159)(160)(161) **(EG 2)** Etiologies may vary (eg, aortic dissection, atherosclerosis, coagulopathy, embolization, mechanical compression, systemic hypoperfusion, vasculitis).(159)(161)(162)(163) **(EG 2)** For patients experiencing spinal infarct or ischemia, timely identification facilitates reperfusion intervention, which may improve outcomes.(160) **(EG 2)**

For spinal cord stimulator placement planning, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Preoperative MRI of the spine is recommended within 12 months prior to the spinal cord stimulator placement to evaluate potential pathology that could alter the diagnosis or negatively impact the outcome and to assess the need for alternative approaches to placement of the stimulator electrodes.(166)(167)(168) **(EG 2)**

For spinal dysraphism, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Spinal dysraphism includes a variety of malformations with incomplete fusion of midline mesenchymal, bony, and neurologic structures; a broad bony defect of the vertebral arch is seen at the level of the malformation.(63)(72)(170) **(EG 2)** Spinal dysraphism may be associated with neurogenic scoliosis. Intraspinal pathology has been reported in 12% to 50% of patients with congenital scoliosis; the MRI findings may include anomalies such as filum lipoma, syrinx, or Chiari malformation.(150)(174) **(EG 2)**

For spinal stenosis of the cervical spine, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A literature review on the imaging of spinal stenosis noted that CT scan, CT myelography, and MRI can all visualize the ventral and dorsal contributions to spinal canal compromise.(180) **(EG 2)**

For spondylolysis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Prior to the development of detectable stress fracture on plain radiograph or CT scan, bone marrow edema of the pars interarticularis may be identified using thin-slice MRI in patients with spondylolysis.(30)(183)(184) **(EG 2)** MRI may also detect associated ligament disruption in traumatic spondylolysis and is used to evaluate neurologic findings, such as radiculopathy, in patients with spondylolysis.(30)(185)(186) **(EG 2)**

For stereotactic spine radiotherapy, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** CT scan and MRI are increasingly used for treatment planning prior to stereotactic spine radiotherapy for spinal metastases.(187)(188)(190)(192)(193) **(EG 2)**

For syringomyelia in the cervical spine, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline indicates that MRI is an appropriate imaging modality for assessment.(20) **(EG 2)** A literature review recommends that when Chiari malformation is suspected on cranial CT scan, MRI of the entire spine is indicated.(195) **(EG 2)**

For torticollis in children, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles indicate that MRI of the brain and cervical spine is an appropriate imaging study in children presenting with torticollis in the absence of trauma, after an initial negative CT scan of the neck and cervical spine. In newborns and infants with suspected congenital torticollis, ultrasound is the preferred initial imaging study.(197) **(EG 2)** MRI is recommended for cases in which atypical features on ultrasound warrant further evaluation.(122)(197) **(EG 2)**

For trauma or fracture of the cervical spine, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRI is helpful in the assessment of acute and subacute spinal cord trauma and can demonstrate ligamentous and cord injury, displaced disk fragments, and intraspinal hematomas not visible on CT scan.(202)(203)(204)(205)(206) **(EG 1)** For blunt cervical trauma, a meta-analysis concluded that cervical spine MRI has a negative predictive value of 100% for spinal cord injury and is especially useful to evaluate trauma patients who are obtunded or have an altered mental status when CT scan is negative or nondiagnostic.(204) **(EG 1)** Systematic reviews of studies of patients with spinal cord injuries who did not have abnormalities on plain x-rays or CT scans have demonstrated abnormal spine MRI findings in 57% of children and 93% of adults.(207)(208) **(EG 1)** A meta-analysis of 1714 clinically unevaluable blunt trauma patients with initial negative CT scans of the cervical spine noted abnormal MRI scans in 271 patients (15.7%); however, only 5 patients (0.3%) required surgical intervention.(209) **(EG 1)** A prospective study of 767 blunt trauma patients who either were clinically unevaluable or had persistent midline cervicalgia after initial negative CT scans of the cervical spine found that 181 (23.6%) had an abnormal cervical spine MRI. Of these 181 patients, 11 underwent cervical spine surgery after review of the MRI results; however, 6 of the 11 had neurologic signs or symptoms, and the authors concluded that the clinical significance of the additional injuries noted on MRI was unclear.(210) **(EG 2)** Retrospective cohort studies of patients with spinal cord

injury without radiographic abnormalities concluded that early cervical spine MRI may help predict neurologic outcomes.(211)(212) **(EG 2)** A retrospective cohort study of 99 patients with cervical fractures determined that MRI detected abnormalities not seen on CT scan in 48% of cases, which led to a change in management in 39% of cases. Factors associated with abnormalities on MRI included age of 61 years or older, neurologic deficit, multiple trauma status, decreased alertness, and cervical spondylosis.(213) **(EG 2)** A prospective cohort study of 53 children younger than 36 months with documented inflicted head trauma found that 15% had associated cervical spine injuries noted on cervical spine MRI.(214) **(EG 2)** In a retrospective cohort study of 206 pediatric patients evaluated for spinal injuries in a trauma center, MRI was obtained more frequently in patients between 0 and 3 years of age (67%) as compared with patients between 4 and 9 years of age (20%). The authors hypothesized that this disparity may be due to difficulty in obtaining reliable neurologic examinations in infants and toddlers. MRI is also a better tool for imaging cartilage, of which a significant portion of the spinal column in this age group is composed.(215) **(EG 2)** A retrospective study of 84 children with cervical trauma found that, when CT scan was used as the standard, MRI identified fractures with sensitivity of 100% and specificity of 97%. The authors recommended that larger, prospective studies be performed to investigate the potential role of MRI in the diagnosis of fractures.(216) **(EG 2)** A case control study compared spine MRI results with intraoperative findings in 35 patients and found that MRI had sensitivity of 80% to 100% and specificity of 64% to 100% for diagnosis of ligamentous, intervertebral disk, and paraspinal muscle injuries.(217) **(EG 2)**

For tuberculosis of the spine, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRI is considered the most accurate imaging modality for diagnosis and classification of tuberculosis, including early spinal cord changes.(88) (243) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(244); 42 CFR 422.101(245)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(246); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(247); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(248)

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Footnotes

[A] Histiocytic and dendritic cell neoplasms are a heterogeneous group of rare hematologic disorders that arise from neoplastic proliferations of cells in the monocytic, histiocytic, or dendritic lineage that arise from common myeloid progenitors.(35)(36) They are classified on the basis of clinical, radiographic, histologic, and molecular features. Major categories include the plasmacytoid dendritic cell neoplasms; Langerhans cell and other dendritic cell neoplasms, including Langerhans cell histiocytosis; and histiocytic neoplasms, including Erdheim-Chester disease and Rosai-Dorfman disease.(35)(36)(37) Depending on the subtype, histiocytic and dendritic cell neoplasms may be characterized by vertebral lesions.(35)(38)(39) [A in Context Link 1]

Definitions

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

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